

Conclusion

This project successfully developed an explanatory and predictive framework to analyze the key drivers of student academic success using the Kaggle *Student Performance Factors* dataset. By combining Exploratory Data Analysis (EDA), data visualization, and Ordinary Least Squares (OLS) linear regression, the analysis uncovered clear, actionable insights into how student habits directly influence exam outcomes.

The final regression model demonstrated a robust fit for behavioral data, achieving an R-squared (R^2) value of 0.3857, meaning the model explains 38.6% of the variance in final exam scores. The remaining variance is attributed to unmeasured environmental or personal factors. Predictions generated by this model maintain a highly reliable accuracy profile, bound by a Root Mean Squared Error (RMSE) of 8.1318 points.

Statistically significant insights ($p < 0.05$) derived from the model include:

- **Academic Investment:** Every additional hour spent studying increases a student's final score by 3.98 points, identifying study time as the single most powerful positive driver of academic performance.
- **Physical Wellness:** Sleep hours and class attendance are vital baseline habits. Each extra hour of sleep boosts scores by 1.08 points, while every 10% increase in attendance yields an estimated 3.3-point increase on the exam.
- **Socioeconomic Trade-offs:** Holding a part-time job imposes a heavy academic penalty, dropping a student's performance by 3.97 points compared to non-working peers. This highlights a critical risk factor for students balancing employment with schoolwork.
- **Insignificant Factors:** Demographics such as parental education level and participation in extracurricular activities showed no statistically verifiable impact on exam scores within this model layout ($p > 0.05$).

Strategic Summary:

Ultimately, this project proves that student success is heavily dictated by controllable behavioral variables rather than static background demographics. Educational institutions and counselors can leverage these specific mathematical slopes to build data-driven intervention strategies—such as monitoring at-risk working students, flagging falling attendance percentages, and proactively promoting balanced sleep and study habits to optimize student grade distributions.